ALGEBRA 2 · UNIT 4: POLYNOMIAL FUNCTIONS

Water Slide Designer

Design a smooth water slide with polynomial zeros, multiplicity, and end behavior.

Honors

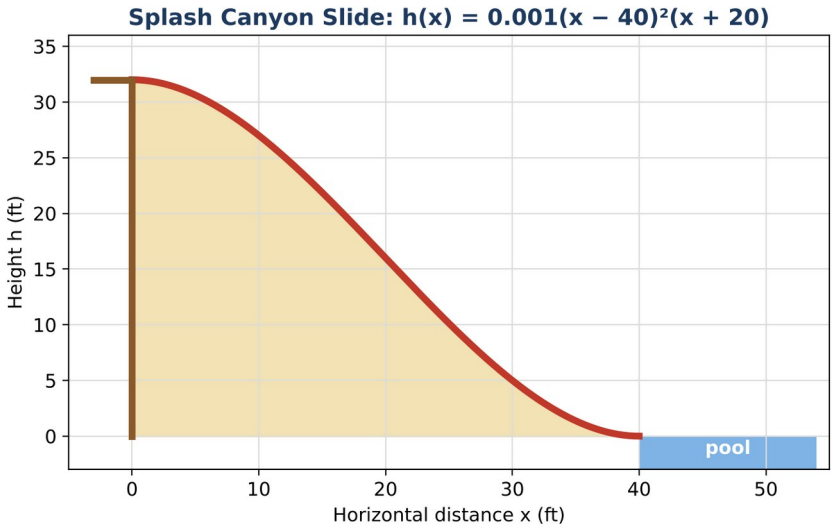
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Splash Canyon Water Park is building a new slide. The profile is $h(x) = 0.001(x - 40)^2(x + 20)$, where x is the horizontal distance in feet and h is the height in feet.

The double zero at x = 40 makes the slide level out smoothly at the pool. You will analyze this design and then create your own.



Learning Goals

- I can derive a general polynomial model from design conditions.
- I can prove properties of the model algebraically.
- I can design under several constraints.

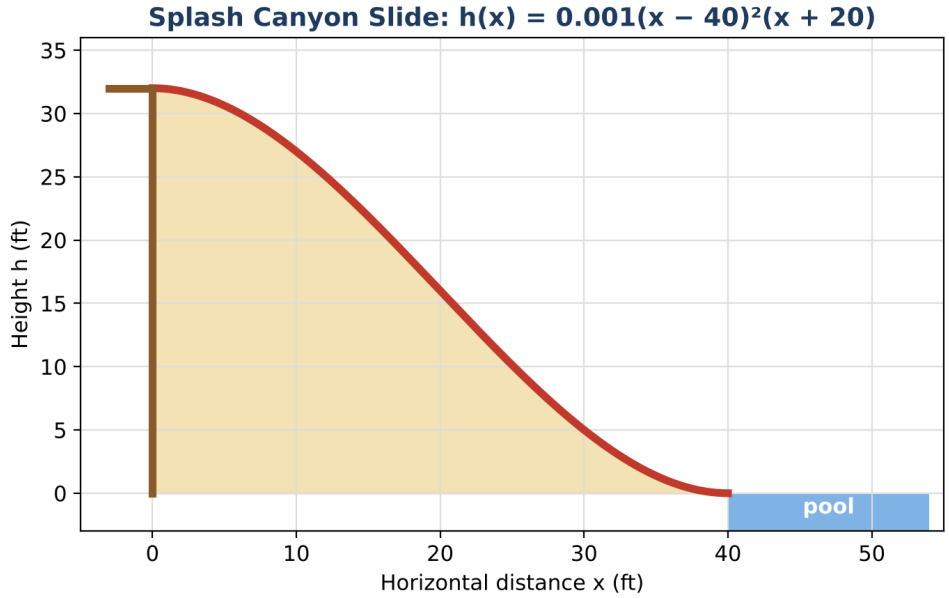
Project Timeline

When	What you will do
Day 1	General Slide Formula
Day 2	Theorems
Day 3	Design Within Constraints
Final	Slide Design Report + reflection

Part 1: General Slide Formula

Day 1 · Slides of the form $h(x) = a(x - L)^2(x + 0.5L)$ start at height H and land at x = L.

1. Show that $h(0) = 0.5aL^3$ and find a in terms of H and L. Check with Splash Canyon (H = 32, L = 40).



2. Expand h(x) and show $h(x) - H = a\,x^2(x - 1.5L)$.

3. Use the identity to prove that the start (x = 0) is the highest point of the slide for $0 \le x \le L$.

Part 2: Theorems

Day 2 · Use $p(x) = x^3 - 60x^2 + 32{,}000$.

4. Use the Rational Root Theorem and synthetic division to factor p completely.

5. Find the remainder when p is divided by $(x - 30)$ and interpret it.

6. Find the average rate of change of h from 0 to 20 and from 20 to 40. Where is the slide steeper?

Part 3: Design Within Constraints

Day 3 · Start height 30 ft; the slide must be at least 12 ft high at x = 15; the landing must be smooth.

7. Find the smallest L that satisfies the constraints (test values or solve).

8. Write the final polynomial and graph it.

9. Write the report with your proofs.

Final Product: Slide Design Report

Your final product must include:

- General formula and proof
- Factoring and rate analysis
- Constrained design
- Report

Reflection

R1. How does a general formula help the park design many slides?

R2. Why is the start the highest point?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original slide polynomial starts at the required height and lands smoothly (double zero), with work shown. Goes beyond the minimum with a creative, realistic choice.	Original slide polynomial starts at the required height and lands smoothly (double zero), with work shown.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes the polynomial in factored and standard form, a graph with zeros and multiplicity, division checks, and a design. Polished, organized, and visually engaging.	Report includes the polynomial in factored and standard form, a graph with zeros and multiplicity, division checks, and a design. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.